INCH-POUND

MIL-B-52817B(ME)
21 May 1991
SUPERSEDING
MIL-B-52817A
20 June 1979

MILITARY SPECIFICATION

BRACKET, SCAFFOLD, FOLDING

This specification is approved for use within the USA Belvoir Research, Development and Engineering Center, Department of the Army, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers a folding scaffold bracket with an attached handrail adapter bracket.

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications, standards, and handbooks. The following specifications, standards, and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation (see 6.2).

SPECIFICATIONS

FEDERAL

FF-B-575 - Bolts, Hexagon and Square.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: USA Belvoir Research, Development, and Engineering Center, ATTN: STRBE-TSE, Fort Belvoir, VA 22060-5606 by using the Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

AMSC N/A FSC 5440

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FF-N-836	- Nut: Square, Hexagon, Cap, Slotted, Castle, Knurled, Welding and Single Ball Seat.
PPP-B-601	- Boxes, Wood, Cleated-Plywood.
PPP-B-621	- Boxes, Wood, Nailed and Lock-Corner.

MILITARY

MIL-T-704	- Treatment and Painting of Materiel.

STANDARDS

MILITARY

MIL-STD-105	- Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-129	- Marking for Shipment and Storage.
MIL-STD-889	- Dissimilar Metals.
MIL-STD-1186	- Cushioning, Anchoring, Bracing, Blocking and Waterproofing; with Appropriate Test Methods.
MS35355	- Bolt, Machine, Square Head, Steel, Cadmium or Zinc Plated, UNC-2A.

(Unless otherwise indicated, copies of federal and military specifications and standards are available from the Standardization Documents Order Desk, Building 4D, 700 Robbins Avenue, Philadelphia, PA 19111-5094.)

2.2 Non-Government publications. The following documents form a part of this document to the extent specified herein. Unless otherwise specified, the issues of the documents which are DoD adopted are those listed in the issue of the DODISS cited in the solicitation. Unless otherwise specified, the issues of documents not listed in the DODISS are the issues of the documents cited in the solicitation (see 6.2).

AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM)

A 53	- Standard Specification for Pipe, Steel, Black and Hot-Dipped, Zinc-Coated, Welded and Seamless.
A 506	- Steel, Sheet and Strip, Alloy, Hot-Rolled and Cold-Rolled, Regular Quality and Structural Quality.

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia PA 19103.)

SOCIETY OF AUTOMOTIVE ENGINEERS, INC (SAE)

SAE J 429	- Mechanical and Material Requirements for Externally Threaded Fasteners.

(Application for copies should be addressed to the Society of Automotive Engineers, Inc., 400 Commonwealth Drive, Warrendale, PA 15096.)

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NATIONAL MOTOR FREIGHT TRAFFIC ASSOCIATION (NMFTA)

National Motor Freight Classification Rules.

(Application for copies should be addressed to the American Trucking Association, ATTN: Traffic Order Section, 2200 Mill Road, Alexandria, VA 22314.)

UNIFORM CLASSIFICATION COMMITTEE (UCC)

Uniform Freight Classification Rules.

(Application for copies should be addressed to the Uniform Classification Committee, ATTN: Tariff Publishing Officer, Room 1106, 222 South Riverside Plaza, Chicago, IL 60606.)

(Non-Government standards and other publications are normally available from the organizations that prepare or distribute the documents. These documents also may be available in or through libraries or other informational services.)

2.3 Order of precedence. In the event of a conflict between the text of this document and the references cited herein, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 Description. The bracket shall be as shown in figures 1 and 2 and as specified herein.

3.2 Material. Material shall be as specified herein. Materials not specified shall be selected by the contractor and shall be subject to all provisions of this specification.

3.2.1 Sheet steel. Sheet steel shall conform to ASTM A 506, sheet, hot rolled, commercial quality, cut edge.

3.2.2 Pipe. Steel pipe shall conform to ASTM A 53, unthreaded with square cut ends.

3.2.3 Bolts. Square head bolts used in the assembly of the folding scaffold bracket shall conform to MS35355-124, zinc plated, material in accordance with FF-B-575. The square head bolts used to assemble the handrail adapter bracket to the folding scaffold bracket shall conform to MS35355-126, zinc plated, material in accordance with FF-B-575.

3.2.4 Nuts. Nuts shall be in accordance with FF-N-836, type I, style 1, .500-13, UNC-2B, carbon steel, grade A, electrodeposited zinc finish.

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3.2.5 Recovered materials. For the purpose of this requirement, recovered materials are those materials which have been collected from solid waste and reprocessed to become a source of raw materials, as distinguished from virgin raw materials. The components, pieces and parts incorporated in the bracket may be newly fabricated from recovered materials to the maximum extent practicable, provided the bracket produced meets all other requirements of this specification. Used, rebuilt or remanufactured components, pieces and parts shall not be incorporated in the bracket.

3.2.6 Material deterioration prevention and control. The bracket shall be fabricated from compatible materials, inherently corrosion resistant or treated to provide protection against the various forms of corrosion and deterioration that may be encountered in any of the applicable operation and storage environment to which the bracket may be exposed.

3.2.7 Dissimilar metals. Dissimilar metals shall not be used in intimate contact with each other unless protected against galvanic corrosion. Dissimilar metals and methods of protection are defined and detailed in MIL-STD-889.

3.2.8 Identification of materials and finishes. The contractor shall identify the specific material, material finish or treatment for use with component and subcomponent, and shall make information available upon request to the contracting officer or designated representative.

3.3 Treatment and painting. The portions of the bracket normally painted shall be cleaned, treated, and painted in accordance with MIL-T-704, type A, color as specified (see 6.2).

3.4 Bracket assembly. The bracket will be assembled as shown in figure 1. The handrail adapter bracket shall not be assembled to the scaffold bracket; however, the 1.25 inch square head bolt with nut will be placed into the 0.5625 inch drilled hole in this bracket prior to shipment.

3.5 Workmanship.

3.5.1 Metal fabrication. Metal used in fabrication shall be free from kinks and sharp bends. The straightening of material shall be done by methods that will not cause damage to the material. Corners shall be square and true. Flame-cutting, using tips suitable for the thickness of the steel, may be employed instead of shearing and sawing. All bends shall be made with controlled means to insure uniformity of size and shape. Precaution shall be taken to avoid overheating. Heated steel shall be allowed to cool slowly. External surfaces shall be free of burrs, sharp edges, and corners, except when sharp edges or corners are required or where they are not detrimental to safety.

3.5.2 Welding. The surfaces of parts to be welded shall be free from rust, scale, paint, grease, mill scale that can be removed by chipping and wire brushing, and other foreign matter. Parent materials, weld filler metals, and fabrication techniques shall be as required to enable the bracket to conform

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to the examination and test requirements specified in section 4. Parts to be joined by fillet welds shall be brought into as close contact as possible and in no event shall be separated by more than 0.187 inch unless appropriate bridging techniques are used. Unless otherwise specified (see 6.2) the welding process used in fabrication of the bracket shall be at the option of the contractor.

3.5.3 Weld quality.

a. The contractor shall be responsible for determining the inspections needed to insure that, when the welded parts are assembled together to make the bracket, the bracket shall conform to the inspection requirements specified in section 4.
b. All weldments shall be free of slag, flux, weld spatter, and other impurities detrimental to either the appearance or strength of the weldment.
c. Undercut in weldments shall be held to a minimum and shall not be more than 0.031 inch deep.

3.5.4 Bolted connections. Boltholes shall be accurately formed and shall have the burrs removed. Washers or lockwashers shall be provided where necessary. Matching thread areas securing bolts conforming to SAE J429 or capscrews shall be of sufficient strength to withstand the tensile strength of the bolt. All fasteners shall be correctly torqued and shall have full thread engagement. Bolts shall protrude not more than 2 full threads.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements (examinations and tests) as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to ensure supplies and services conform to prescribed requirements.

4.1.1 Responsibility for compliance. All items shall meet all requirements of sections 3 and 5. The inspection set forth in this specification shall become a part of the contractor's overall inspection system or quality program. The absence of any inspection requirements in the specification shall not relieve the contractor of the responsibility of ensuring that all products or supplies submitted to the Government for acceptance comply with all requirements of the contract. Sampling inspection, as part of manufacturing operations, is an acceptable practice to ascertain conformance to requirements, however, this does not authorize submission of known defective material, either indicated or actual, nor does it commit the Government to acceptance of defective material.

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to requirements, however, this does not authorize submission of known defective material, either indicated or actual, nor does it commit the Government to acceptance of defective material.

4.1.2 Component and material inspection. The contractor is responsible for ensuring that components and materials are manufactured, examined, and tested in accordance with referenced specifications and standards, as applicable.

4.2 Classification of inspections. The inspection requirements specified herein are classified as follows:

a. Quality conformance inspection (4.3).
b. Inspection of packaging (see 4.5).

4.3 Quality conformance inspection.

4.3.1 Sampling. Sampling for examination shall be in accordance with MIL-STD-105. Sample size shall be determined by table I and table IIa, inspection level II or S-3.

4.3.2 Examination.

4.3.2.1 Samples. Samples selected in accordance with 4.3.1 shall be examined for the major and minor defects specified in 4.4.1. Presence of one or more defects shall be cause for rejection.

4.4 Inspection procedure.

4.4.1 Examination. The bracket shall be examined as specified herein for the following defects:

Major

101. Bracket not as shown in figures 1 and 2 (see 3.1).
102. Material not as specified (see 3.2).
103. Sheet steel not as specified (see 3.2.1).
104. Pipe not as specified (see 3.2.2).
105. Bolts not as specified (see 3.2.3).
106. Nuts not as specified (see 3.2.4).
107. Used, rebuilt or remanufactured components, pieces, or parts incorporated in the bracket (see 3.2.5).
108. Materials are not resistant to corrosion or deterioration or treated to be made resistant to corrosion or deterioration for the applicable storage and operating environment as specified (see 3.2.6).
109. Dissimilar metals as specified in MIL-STD-889 are not effectively insulated from each other as specified (see 3.2.7).
110. Contractor does not have documentation available for identification of material, material finishes, or treatments (see 3.2.8).
111. Workmanship not as specified (see 3.5).

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Minor

201. Treatment and painting not as specified (see 3.3).
202. Bracket assembly not as specified (see 3.4).

4.5 Inspection of packaging.

4.5.1 Quality conformance of packaging.

4.5.1.1 Unit of product. For the purpose of inspection, a completed pack prepared for shipment shall be considered a unit of product.

4.5.1.2 Sampling. Sampling for examination shall be in accordance with MIL-STD-105. Sample size shall be determined from table I and table IIa.

4.5.1.3 Examination. Samples selected in accordance with 4.5.1.2 shall be examined for the following defects. Presence of one or more defects shall be cause for rejection.

112. Packing not as specified (see 5.1).
113. Marking not as specified (see 5.2).

5. PACKAGING

5.1 Packing. Packing shall be level A, B, or C as specified (see 6.2).

5.1.1 Level A. The brackets shall be folded for minimum cubage and with the handrail adapter brackets, shall be packed in boxes conforming to PPP-B-601, overseas type, or PPP-B-621, class 2. Cushioning shall conform to MIL-STD-1186 to prevent movement and resulting damage to the painted surfaces.

5.1.2 Level B. The brackets shall be packed as specified in 5.1.1 for level A, except the boxes shall be domestic type or class as applicable.

5.1.3 Level C. The brackets shall be packed to assure carrier acceptable and safe delivery to destination at lowest rates in compliance with Uniform Freight Classification rules or National Motor Freight Classification rules.

5.2 Marking. In addition to any special marking specified in the contract or purchase order (see 6.2), marking shall be in accordance with MIL-STD-129.

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

6.1 Intended use. The bracket is intended for use in erecting bolted steel tanks. The handrail bracket, which is a component of the scaffold bracket, is intended to serve as a mounting for handrail upright supports. NSN 5440-00-319-0118 and NSN 5440-00-510-0327 which are used in Equipment Set 5180-90-CL-N33 apply thereto.

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6.2 Acquisition requirements. Acquisition documents should specify the following:

a. Title, number, and date of the specification.
b. Issue of DODISS to be cited in the solicitation, and if required, the specific issue of individual documents referenced (see 2.1.1 and 2.2).
c. Color required (see 3.3).
d. Welding process required, when other than at the option of the contractor (see 3.5.2).
e. Level of packing required (see 5.1).
f. Any special marking (see 5.2)

6.3 Changes from previous issue. Marginal notations are not used in this revision to identify changes with respect to the previous issue due to the extensiveness of the changes.

6.4 Subject term (key-word) listing.

Folding scaffold bracket
Handrail adapter bracket
Scaffold bracket

Custodian:
Army - ME

Review Activity:
DLA - CS

Preparing activity:
Army - ME

Project 5440-A013

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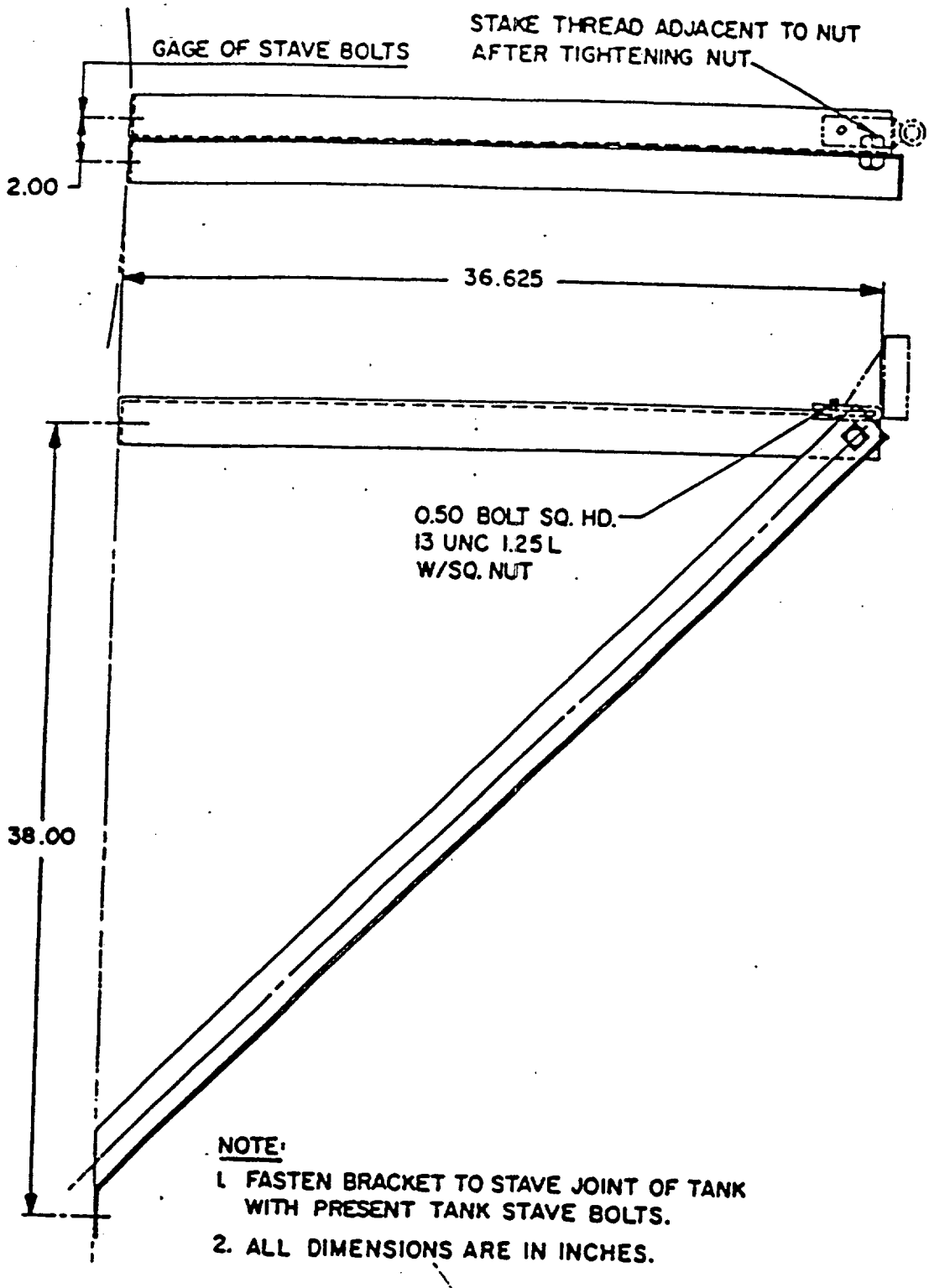


FIGURE 1. Folding scaffold bracket assembly.

X-2899A

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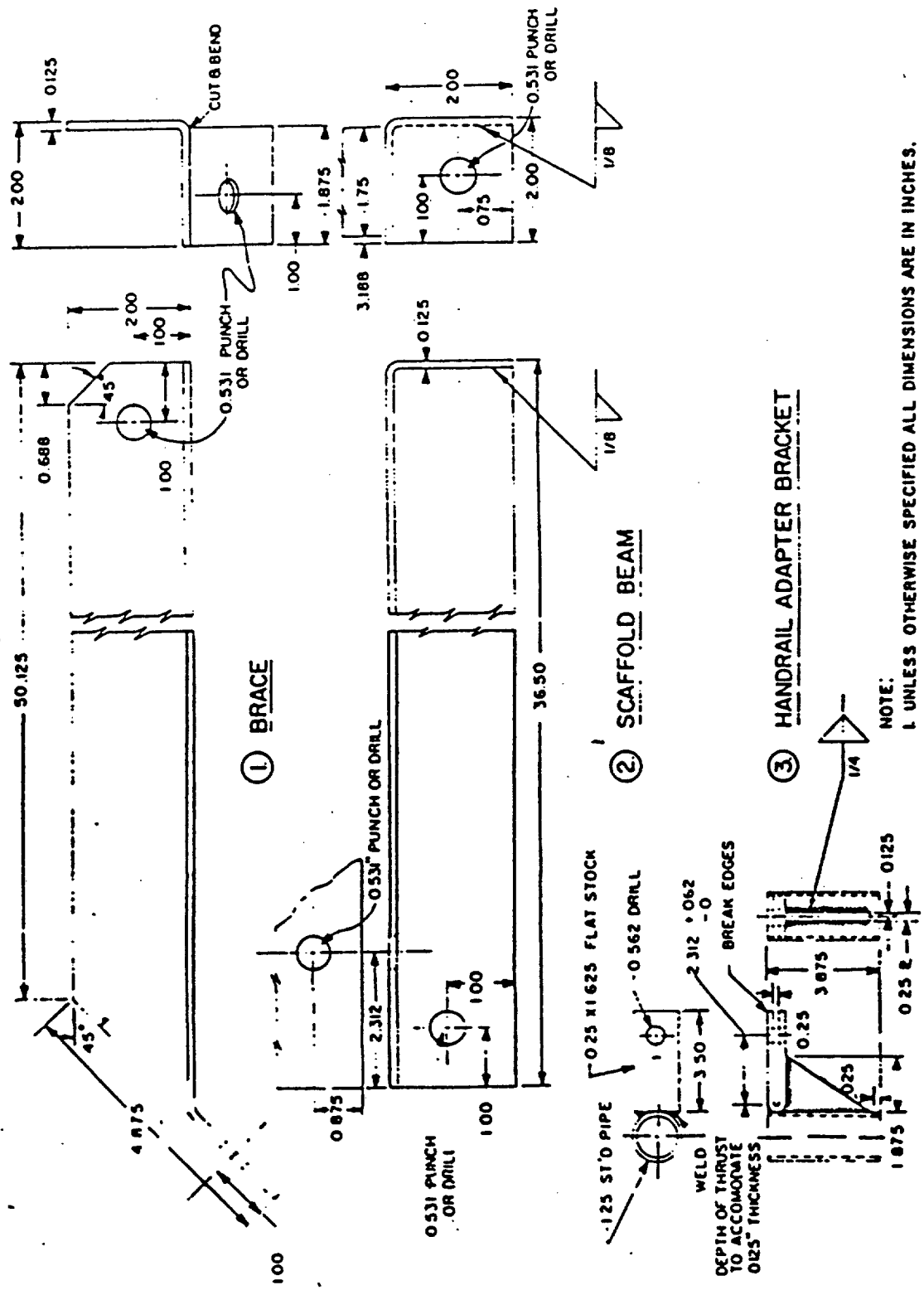


FIGURE 2. Folding scaffold bracket, details.

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INSTRUCTIONS

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2. The submitter of this form must complete blocks 4, 5, 6, and 7.
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NOTE: This form may not be used to request copies of documents, nor to request waivers, or clarification of requirements on current contracts. Comments submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or to amend contractual requirements.

I RECOMMEND A CHANGE:	1. DOCUMENT NUMBER MIL-B-52817B(ME)	2. DOCUMENT DATE (YYMMDD) 910521

3. DOCUMENT TITLE
Bracket, Scaffold, Folding

4. NATURE OF CHANGE (*Identify paragraph number and include proposed rewrite, if possible. Attach extra sheets as needed.*)

5. REASON FOR RECOMMENDATION

6. SUBMITTER

a. NAME (Last, First, Middle Initial)	c. ORGANIZATION	
b. ADDRESS (Include Zip Code)	d. TELEPHONE (Include Area Code) (1) Commercial (2) AUTOVON (If applicable)	7. DATE SUBMITTED (YYMMDD)

8. PREPARING ACTIVITY

a. NAME	b. TELEPHONE (Include Area Code) (1) Commercial	(2) AUTOVON
	(703) 664-5717	354-5717
c. ADDRESS (Include Zip Code) US Army Belvoir RDE Center ATTN: STRBE-TSE Ft. Belvoir, VA 22060-5606	IF YOU DO NOT RECEIVE A REPLY WITHIN 45 DAYS, CONTACT: Defense Quality and Standardization Office 5203 Leesburg Pike, Suite 1403, Falls Church, VA 22041-3466 Telephone (703) 756-2340 AUTOVON 289-2340	

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